prompt and cause misinterpretation and incorrect calculation. For example, a question like "What is half of the sum of 20 and 30?" could be misinterpreted, resulting in an incorrect calculation if the model fails to recognize that it should first calculate the sum before halving it based on the rules of math.

- *Model limitations*—Despite advancements in AI and natural language processing, models like GPT-4 are not perfect. They are trained on vast datasets and can generate sound responses, but they do not possess true understanding or reasoning capabilities. Therefore, they might hallucinate parts of their answers or just provide false statements. For example, sometimes the AI model fails to count the specific number of characters in a word, leading to inaccuracies.
- *Overfitted examples*—If the model is provided with examples that are overly specific or not representative of the broader problem space, it may "overfit" to those examples. For instance, if the examples only cover problems with single-digit numbers, the model may struggle when presented with larger numbers or more complex expressions. This can result in the model applying the wrong logic or methodology when faced with a slightly different problem.

For all of these reasons, it is essential to validate the correctness of each step in the reasoning process, especially for complex calculations or tasks that require precise answers. This validation is crucial because even though chain-of-thought prompting encourages the model to articulate its reasoning, it does not guarantee that every step will be accurate.

However, chain-of-thought prompting is a very accessible technique since it does not require us to fine-tune the AI model or undergo complex training processes. This accessibility is one of the key advantages of using chain-of-thought prompting in various applications.

5.3.2 General methodology for creating chain-of-thought prompts

Just as with few-shot learning methodologies, developing effective chain-of-thought prompts requires a systematic approach. After all, we are not called "engineers" without a reason! The following methodology focuses on guiding LLMs to explain their reasoning process clearly and logically, aiming for accurate and reliable results. Figure 5.7 is a visualization of the general steps involved for crafting prompts using chain-of-thought reasoning as an example.

Here are the general steps involved:

1. *Define the structure of the prompt.* Begin by establishing a clear format for the prompt that outlines the AI's expected response. By providing a well-structured template, you can ensure that the model understands the required information and the format in which it should be presented. For example, "Calculate the area of a circle: Area = $\pi * R^2$."

   In this example, the prompt states the task at hand (calculating the area of a circle) and provides the formula to be used. This structure sets the stage for the model to follow a specific pattern when generating its output.